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MicroPack-HE: A Packing-Aware Lightweight Homomorphic Encryption Framework for Secure Analytics in Resource-Constrained IoT Networks

Abstract- The explosive growth of Internet of Things (IoT) devices has further highlighted the requirement to safely perform data analytics over resource-constrained networks, for which existing encryption primitives are insufficient for supporting in-network computation and traditional homomorphic encryption (HE) is not practically computable. Recently-proposed light-weight HE schemes trade-off either security assurances or architectural overhead by employing adaptive, federated or multi-stage constructions and thus lack deployability on low-power computation devices. To mitigate these challenges, we propose the MicroPack-HE, a new packing-aware lightweight homomorphic encryption framework for resource-limited IoT systems in this work. MicroPack-HE relies on a constraint additive core making it simple to handle first-order aggregation tasks as well as controlling such computational and memory overhead. The main novelty contributes to a fixed-size plaintext packing method, where multiple low-bit sensor samples are packed together into one ciphertext by an efficient bit-level tiling approach, and thus greatly reduces encryption times of the device and the communication cost. In contrast to FHE schemes, MicroPackHE purposefully limits the homomorphic expressiveness such that it only supports addition and one small scaling multiplication for efficient running on 32-bit microcontrollers without specialized hardware. Computationally intensive assistance components are pre-computed and stored at the cloud side, so that the IoT devices do the negligible arithmetic operations in encryption. Experimental results on ESP32-class hardware show that MicroPack-HE reduces encryption latency by 41.6% compared to the off-the-shelf Paillier-based ones and energy demand is 38.2% less per transmission. Effective packing helps save 57.4% the size of ciphertext and network overhead. However, with these improvements the framework remains semantically secure under standard additive HE assumption and accomplishes data integrity without expensive message authentication. The results confirm that MicroPack-HE is a feasible and efficient privacy-preserving analytics approach for emerging IoT networks.

Keywords- Lightweight homomorphic encryption, IoT security, resource-constrained networks, ciphertext packing, privacy-preserving analytics, secure data aggregation

I Introduction

The explosive growth of Internet of Things (IoT) has turned the traditional data collection model into these highly distributed, data-intensiveness ecology [1][2], which can enable large-scale sensing, monitoring, and analytics at network edge. The present-day IoT deployments cover wide range of application domains like smart cities [3][4], industrial automation, health monitoring and intelligent traffic system [5], where millions to billions of low-powered sensor nodes incessantly stream sensitive data. Aiming at reducing latency, decreasing bandwidth consumption and alleviating the reliance on cloud computing [6][7], edge-assisted and in network analytics has become extremely important. Nevertheless, such a

migration brings with it huge trust and security issues: raw sensor data are not rarely streamed over untrusted networks or collected by semi-trusted aggregation nodes. Preserving the privacy of data, while keeping the possibility of meaningful computation not a trivial task and poses some fundamental questions on resource-limited environments as an IoT device. Existing encryption techniques, although widely adopted to secure data at rest and during transmission, naturally preclude computation over encrypted values, requiring decryption at intermediate nodes, and hence exposure secrets [8][9]. Homomorphic encryption (HE) can provide a practicable approach since it allows to directly compute over ciphertexts, but the most schemes for HE or even fully homomorphic encryption (FHE) that have been proposed so far involve extremely high computational, memory and energy costs. Such schemes rely upon extensive integer arithmetic, intricate noise control, and multiple ciphertext operations which are not feasible for the low-end microcontrollers found in IoT devices [10][11]. All recent existing light-weight or partially homomorphic solutions even, unnecessarily bring architectural complexity, via adaptive protocols, federated learning paths [8][9], multi-stage encryption etc. reducing the practicality and deployability on resource and communication constrained networks [12].

A crucial but still widely unaddressed vacuum is the need for packing aware: lightweight homomorphic encryption solutions tailored to abide by the data properties and operational needs of IoT sensing workloads. In most real-world applications, sensor readings are low-bit signals that may be collected periodically [13] and naturally lend themselves for aggregation such as addition or averaging. Currently, most of the lightweight HE schemes only encrypt per reading independently and the frequency of encryption is too high that lead to over-inflated communications overheads and superfluous energy consumption. Furthermore, most designs favour cryptographic generality over system efficiency [14], and do not take advantage of fixed-size data representation, bounded-arithmetic and asymmetric workload distribution between devices and the cloud. At present, however, there does not exist a publicly available practical HE framework that is able to jointly optimize computation complexity, ciphertext compactness and deployability onto commodity IoT hardware. To overcome these drawbacks, in this paper, we present MicroPack-HE, which is a packing-aware light-weight homomorphic encryption (LWHE) system suited for secure analytics on resource-constrained IoT networks. MicroPack-HE is designed primarily to support efficient private data aggregation by using a bounded additive homomorphic encryption (BAHE) core combined with a fixed-size plaintext packing approach, which packs multiple low-bit sensor readings into one ciphertext. By deliberately constraining homomorphic expressiveness to addition and a single delicate scalar multiplication, as well as offloading computational expensive mechanisms onto the cloud via precomputation, MicroPack-HE can achieve low device-side overheads while maintaining semantic security under standard additive HE assumptions [15][16].

The key objectives are: (i) to minimize the latency and energy consumption of encryption operations on microcontroller class devices, (ii) to reduce communication overhead via a lightweight ciphertext-packing scheme, and (iii) to substantiate the feasibility of lightweight homomorphic analytics through real-device experiments. The performances demonstrate that

MicroPack-HE is a scalable and deployable approach for privacy-preserving analytics in future IoT systems.

The rest of this paper is structured as follows. We summarize in Section II the related work including the lightweight cryptography and homomorphic encryption for resource-constrained IoT scenario, and point out the research gap that we fill. Section III describes the system model, threat model and security assumption. Section IV describes the design of the proposed MicroPack-HE, which incorporates the bounded additive homomorphic encryption core, plaintext packing scheme, and processing mechanisms (encryption/decryption operations, cloud-side precomputation), followed by complexities analysis. Experimental setup and performance evaluation procedures are explained in Section V. Section VI presents the experimental results, comparison with other methods and ablation study. Finally, the last section concludes the paper and points out possible future research directions.

II Literature Review

Kuldeep, Zhang Dual (2021), problem on data privacy and communication efficiency in internet of things (IoT) was studied via compressive sensing (CS) as a joint compression-encryption scheme. Albeit promising for massive sensor data processing, even computationally-secure CS schemes turn out to be susceptible to ciphertext-only attacks under short signal length due their own recovery conditions, which is common in the case of IoT sensing as in [1]. They addressed this challenge by introducing an energy concealment (EC) based encryption scheme which has the property of providing perfect secrecy and does not rely on an additional secure key delivery channel. Even though their solution reduced energy consumption and memory footprint significantly in comparison to AES on TelosB motes, it does not support computation over encrypted data which limits the application of storing encrypted primitives for secure in-network analytics. Palaparthi et al. (2022) [2] moved to attacking protocol-level lightweight security, conducted a comparative evaluation of IoT security protocols including LSMWP, CoAP with DTLS and efficient cryptographic primitives for constrained devices. They performed a study of the cryptographic strength, key management resilience, integrity-protection and processing overhead in their work. Although the authors present very interesting trade-offs between security and efficiency, the considered protocols are intrinsically communication-oriented ones, that do not take into account privacy-preserving data processing or encrypted aggregation concepts which are a must-have feature in today's IoT-driven analytics. Thakor et al. (2021) [3] presented a detailed comparison on lightweight cryptographic algorithms developed for IoT devices with limited resources and compared over fifty existing schemes and the most recent NIST new candidates. The analysis performed by them focused on the implementation cost as well as the software and hardware performance, resistance against cryptanalytic attacks. The authors pointed out that, while lightweight cryptography can save a lot of resources in real applications, most schemes only consider the aspects of data privacy and authentication with little or no attention to secure computation or analytics over encrypted data. Ding et al. (2021) [4] focus on IoT security in terms of authentication and presented a lightweight authentication scheme for anonymous entities in the IoT system with elliptic curve

cryptography (ECC) and signcryption. Their scheme lowered the computational and communication costs with a provable security guarantee under the RO model. Despite these benefits, identity authentication is the only secure operation provided by the work and no integration with encrypted data aggregation and privacy preserving computation, which are two functionalities that are more in demand when it comes to IoT based systems for big data. Aljaedi et al. (2025) [5], explored lightweight encryption techniques from the perspective of an unholy trinity that combined quantum encryption ideas, chaotic maps, and metaheuristic learning. Their method had excellent statistical security characteristics and was robust against many attack models. Yet, it is tailored for multimedia content encryption as opposed to low-bit sensor readings, and its computationally expensive nature makes it infeasible for real-time analytics or homomorphic computation on microcontroller-class IoT devices. Li et al. (2021) [6] were one of the first who brought homomorphic encryption to IoT environments and discussed a lightweight privacy-preserving protocol in industrial IoT applications. Under their approach, the use of a user-preserving protocol to perform cryptographic computation over encrypted data in untrusted cloud servers was made possible. While few computational optimisations were proposed, however the scheme comes at a non-trivial overhead and does not take benefit of data packing or bound arithmetic to scale well on tight edge devices.

Sheela et al. (2025) [7] Generalized the concept of homomorphic encryption to reinforcement learning centered WSN's via combining HE with Quantum-safe cryptography for the purpose of model parameter aggregation. The framework has the extremely good privacy pros against the attackers and a high network gap, but it comes at a cost of significant architectural complexity involving cooperative decryption and an iterative learning cycle which becomes unsuitable for elementary type sensing nodes having severe energy and memory limitation. Ali et al. (2023) [8] combined homomorphic encryption and block chain technology for better privacy preservation in IoT based health system. Theirs system supported encrypted computation, fine-grained access control and auditability using smart contracts. Even though it is resilient, the aggregated overhead of blockchain and HE makes deployment possible only on moderately powerful nodes; low-power IoT sensors are hence prevented from being involved in encrypted analytics directly. Mohammed et al. (2025) which is a taxonomy based survey on lightweight cryptographic schemes for industrial IoT providing a classification of the schemes based on algorithmic type, deployment situation and performance attributes [9]. Their work also stressed the increase in demand towards hybrid and energy-aware secure solutions while it reveals a dearth of scalable cryptographic constructions that trade-off computation, communication, and security for heterogeneous IIoT environments. Zhang et al. (2024) [10] introduced an anonymous data sharing scheme for resource-limited IoT devices by utilizing elliptic curve operations and blockchain based integrity checking. The comparative advantage of the scheme is that it is more efficient in terms of computation costs with privacy and revocation using elliptic curve instead to bilinear pairings. But, as with other data-sharing platforms, its focus is primarily on secure transmission and verification, not encrypted data analytics. Active and passive homomorphic encryption integration with federated learning for smart cities was addressed by Alqazzaz (2025) [11], they proposed an optimization technique to

minimize latency and overhead communication cost. Although it works at system level, the strategy assumes relatively powerful edge nodes and is not applicable to microcontroller-level IoT things with respect to iterative model updates and encryption-heavy workloads. Radhakrishnan et al. [12] ran the AES-128, SPECK and ASCON algorithms on constrained IoT boards in a comparative analysis of lightweight cryptographic algorithms. They concluded that while some algorithms obtain a favorable performance, all the approaches are essentially non-homomorphic and require decryption prior to processing -at least partially- sensitive data during computation.

Ali & Anwer (2025) [13] had developed a secure healthcare system based on homomorphic elliptic curve cryptography to encrypt the data sent from for wearable sensors. Their construction showed better computational efficiency than that of traditional public-key systems. However, the encryption and decryption latencies still are too high for continuous sensing applications, and no packing mechanism is adopted to decrease transmission overhead. Thabit et al. (2022) [14] proposed a dual layer lightweight homomorphic cryptosystem for cloud computing that makes use of symmetric encryption and multiplicative homomorphism. The approach enhances security and execution time compared to classical cloud cryptosystems, however it is targeted for the cloud environment and disregards the limited hardware of IoT end devices. Halder and Newe (2022) [15] introduced SmartCrypt - a symmetric homomorphic encryption technique for storage and analytics of secure IIoT timeseries data. Their system showed significant improvements in query processing and scalability. However, SmartCrypt targets cloud-assisted architectures and does not specifically minimize encryption frequency or ciphertext size at the sensor level. Amgbara et al. [16] surveyed lightweight machine learning models for personal IoT security, and discussed the tradeoff between model complexity, accuracy and resources. Even though its main focus lays in anomaly detection and intrusion prevention, there are still existing unsolved issues regarding data privacy and encrypted processing that are beyond the reach of a purely ML-based only approach. Hendaoui et al. (2025) [17] introduced EPAD, a PP anomaly detection method that merges optimized homomorphic encryption with federated learning. While EPAD greatly reduces computational cost compared with the traditional HE based IDS systems, it still relies on federated learning facilities and encrypted model exchanges which only holds for ultra-limited IoT nodes. Finally, Hsu et al. (2023) [18] focused on secure group communication in Internet of Medical Things and presented a lightweight group key agreement protocol using XOR operation and symmetric polynomial. Their scheme acquires low computational and communication overhead, but it only supports secure key establishment without the support of encrypted data aggregation and computation.

2.1 Research Gap

Overall, the related works show that there has been a great development in lightweight cryptography (PRF), authentication and privacy preserving IoT protocols. Nevertheless, existing works either (i) do not enable computation over encrypted data, (ii) are based on heavy homomorphic or federated architectures, or (iii) do not consider optimizing the computation results for low-bit sensor workloads as well as for reducing the size and

frequency of encryption operations. This gap in the state-of-the-art supports the justification for a packing-aware, bounded homomorphic encryption scheme which allows efficient secure aggregation in a way which can be realized on microcontroller-class IoT devices and this is precisely the goal achieved by our MicroPack-HE framework.

III System and Threat Model

The system follows a three-tier IoT architecture consisting of resource-constrained sensing devices, aggregation gateways, and a cloud-based analytics server. IoT devices are assumed to be ESP32-class microcontrollers with 32-bit processors, limited memory, and strict energy constraints, capable of performing only lightweight arithmetic operations without cryptographic acceleration [19][20]. Each device periodically collects low-bit sensor readings and encrypts them locally before transmission. Gateways forward and aggregate encrypted data without accessing plaintext, while the cloud performs homomorphic aggregation and supports cryptographic precomputation. The adversary is modeled as honest-but-curious, where gateways and cloud servers follow protocol specifications but attempt to infer sensitive information from observed ciphertexts or aggregated outputs. The attacker may passively eavesdrop on communication channels but is not assumed to compromise device hardware or secret keys. The primary security goals are to ensure confidentiality of individual sensor readings during transmission and aggregation [21][22], and to maintain data integrity through lightweight authentication mechanisms. Under these assumptions, MicroPack-HE enables efficient privacy-preserving data analytics while adhering to the strict resource limitations of IoT devices.

3.1 Preliminaries

3.1.1 Additive Homomorphic Encryption

Additive Homomorphic Encryption (AHE) allows the addition of ciphertexts representing plaintexts and for the decryption of the result in-order to be equal to a sum of plaintext values. Formally, for an encryption function $Enc(\cdot)$ and a decryption function $Dec(\cdot)$, an AHE scheme meets the following criteria:

$$Dec(Enc(m_1) \oplus Enc(m_2)) = m_1 + m_2 \quad (1)$$

In Eq. 1, m_1 and m_2 are in the clear text and represent a cryptography operation. Some, (but not all) such schemes also allow an encrypted value to be multiplied by a small public scalar. Unlike fully Homomorphic Encryption (FHE) [23][24], AHE does not require costly noise management and ciphertext refresh operations, which is more suitable for energy and resource-constrained IoT devices. MicroPack-HE leverages this bounded additive homomorphic property to enable secure aggregation of sensor reading and uniformly limit the set of allowed operations in order to decrease computational and storage overhead.

3.1.2 Security Assumptions

Security definition for MicroPack-HE MicroPack-HE assumes semantic security in the standard model of additive homomorphic encryption, ciphertexts are indistinguishable under chosen-plaintext attacks (IND-CPA). The framework runs in an honest-but-curious adversarial setting, where intermediate gateways and cloud servers adhere to protocol

specifications but can try to learn information from encrypted data [25]. We do not assume there is any collusion between the cloud and compromised devices, and that private keys are protected in IoT devices. Integrity in LS-AP encourages keeping lightweight message authentication codes so that ciphertext tampering and replays with respect to the homomorphic engine is detectable.

3.1.3 Microcontroller Constraints

The intended deployment targets for MicroPack-HE are microcontroller devices in the ESP32 class with tight RAM and flash constraints, and strict energy requirements. Such devices have no hardware support for large integer arithmetic and cryptographic acceleration, and consequently conventional public-key encryption or fully homomorphic encryption are impracticable. Therefore, encryption operations should be lightweight, in terms of computations, memory occupancy and execution time predictability [26][27]. These limitations motivate the use of fixed-size plaintext representations, bounded arithmetic and a packing-aware design that aims to reduce the amount of times encryption is required and the communication overhead.

3.1.4 Notation Table

Table 1: Notation Used in MicroPack-HE

Symbol	Description
m	Plaintext sensor reading
M	Packed plaintext containing multiple readings
c	Ciphertext
$Enc(\cdot)$	Encryption function
$Dec(\cdot)$	Decryption function
$\oplus$	Ciphertext-domain additive operation
k	Number of sensor readings packed into one plaintext
λ	Security parameter

Table 1 presents mathematical symbols and cryptographic notations employed in this paper to describe the MicroPack-HE framework. Since the presented solution includes additive homomorphic encryption, plaintext packing and resource constrained system, a large number of symbols appear again throughout the design construction, security analysis as well as performance calculation. An explicit notation table is given in this paper for readability purpose and also to avoid ambiguities of symbols, where the same symbol may be different from that according to a value assigned by the readers' background either on cryptography or IoT systems.

4. MicroPack-HE Framework Design

4.1 Design Overview

The Micropack-HE is click-mining oriented, and has been 8202built to be privacy preserving data aggregation tool for IoT type of environments which involves sensors working in ultra constrained computational, memory and energy circumstances [28][29]. In contrast with traditional lightweight cryptosystems that concentrate only on secure transmission and provide no support for encrypted computation [2], or homomorphic frameworks which introduce significant overhead due to high arithmetic complexity and architectural layers [6], MicroPack-HE follows a minimalistically-computation-aware design philosophy. The approach is based on the bounded additive homomorphic encryption core which only allows addition and a small scalar multiplication, and does not require a fully homomorphic expressiveness otherwise imposed by FHE-based approaches [14], thus avoiding the computational cost of such fully expressive schemes. In addition to minimising the encryption frequency and communication overhead, MicroPack-HE makes use of fixed-sized plaintext packaging scheme where multiple low-bit sensor readings are combined into one plaintext based on bit-level tiling technique, selected due to less-entropy of periods and typicality representation in IoT data streams [30]. We intentionally offload computational overhead key-related precomputations and parameter setup to the cloud layer, as it has been observed that the use of cloud can greatly increase system efficiency while maintaining security assumptions [15]. On the device side, encryption uses exclusively simple modular arithmetic that runs on ESP32-type micro-controller which provides predictable time of execution and modest energy requirements. With concerted efforts in the optimization of cryptographic expressiveness, packing efficiency and workload balance, MicroPack-HE strikes a realistic point between security, deployability and practicality that has not been achieved by previous lightweight encryption [3] or homomorphic IoT framework [7]. The architecture diagram for the proposed model is given in Figure 1.

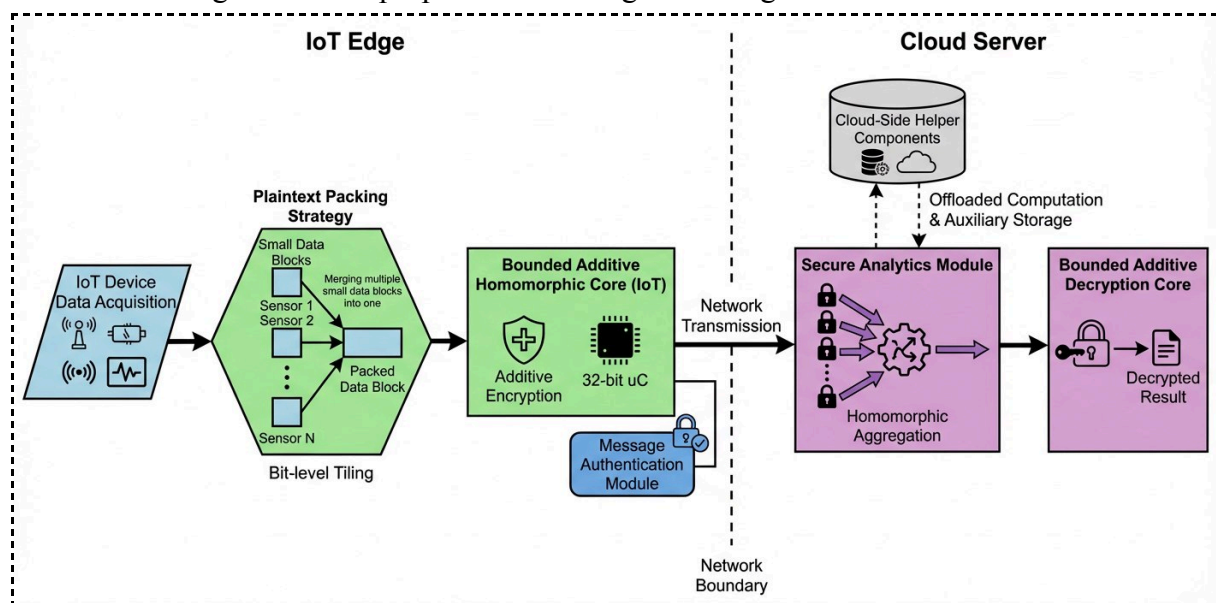


Figure 1. System Architecture of the Proposed MicroPack-HE Framework

4.2 Bounded Additive Homomorphic Encryption Core

At the core of MicroPack-HE lies a bounded additive homomorphic encryption (AHE) construction deliberately tailored to the arithmetic limits and execution capabilities of ESP32-class IoT devices [31]. Let the plaintext space be defined over integers modulo p ,

where p is a fixed public modulus chosen such that $p > \sum_{i=1}^k 2^{b_i}$, ensuring correctness under plaintext packing. For a packed plaintext $M \in \mathbb{Z}_p$, encryption is defined as in Eq.2:

$$c = \text{Enc}(M) = (M + r \cdot p) \bmod N \quad (2)$$

Where r is a fresh random nonce sampled uniformly from a bounded domain and N is a public composite modulus satisfying $N \gg p$. Decryption is performed as in Eq.3:

$$\text{Dec}(c) = c \bmod p \quad (3)$$

which correctly recovers M as long as the accumulated ciphertext-domain operations do not cause modular overflow beyond the bound imposed by N . Given two ciphertexts $c_1 = \text{Enc}(M_1)$ and $c_2 = \text{Enc}(M_2)$, the additive homomorphic property follows directly:

$$\text{Dec}(c_1 + c_2 \bmod N) = (M_1 + M_2) \bmod p \quad (4)$$

MicroPack-HE further supports multiplication of a ciphertext by a small public scalar $\alpha \in \mathbb{Z}$, such that,

$$\text{Dec}(\alpha \cdot c \bmod N) = (\alpha M) \bmod p \quad (5)$$

so long as $(\alpha \cdot M)$ is lower-bounded. In contrast to full homomorphic encryption (HE) schemes calling for expensive noise tracking and bootstrapping operations [14], the MicroPack-HE core bounds the number of required additions and scalar multiplications most strictly while maintaining correctness, such that arithmetic is restricted primarily to lightweight modular addition and multiplication suitable for fast processing on 32-bit microcontrollers [32]. This restricted design conforms to the aggregation-driven nature of IoT analytics workload, which is dominated by functions like summation and averaging [6], and eschews its overly unwieldy generality as in federated or learning-based HE frameworks [7]. With explicit restrictions on homomorphic depth and plaintext magnitude, MicroPack-HE ensures predictable execution time, limited ciphertext expansion and small memory footprint of the homomorphic core that is secure in the standard IND-CPA sense, efficient for resource-constrained IoT networks.

4.3 Plaintext Packing Strategy (Bit-Level Tiling)

However, the MicroPack-HE uses a fixed-size plaintext packing technique with bit-level tiling to pack multiple low-bit sensor readings into one plaintext for encryption, which minimizes the number of encryption operations during communication. Let each sensor of an IoT device produces a stream of k measurements $\{m_0, m_1, \dots, m_{k-1}\}$, each measurement $m_i \in \mathbb{Z}_{2^b}$ is encoded with at most b bits due to the reduced dynamic range of common sensors in widespread use. The packed plaintext M is generated as presented in Eq. 6:

$$M = \sum_{i=0}^{k-1} m_i \cdot 2^{ib} \quad (6)$$

which corresponds to a deterministic bit-level concatenation of individual readings within a single integer. The packing factor k and bit-width b are selected such that,

$$M < p \text{ and } k \cdot b \leq \lceil \log_2 \rceil, \quad (7)$$

such that the packed values in consecutive rounds do not overlap, and decryption is carried out correctly under a modulo boundary [33]. After this aggregation homomorphically, the cloud or e.g., the authorized decryptor retrieves the aggregated packed plaintexts M' from which each respective aggregated readings m'_i is enabled to be extracted through Eq. 7:

$$m'_i = \left\lfloor \frac{M' \bmod 2^{(i+1)b}}{2^{ib}} \right\rfloor \quad (8)$$

For additive aggregation across n devices, correctness is preserved as long as,

$$\sum_{j=1}^n m_{i,j} < 2^b \quad (9)$$

or in general as long as the maximum expected sum falls within the number of bits assigned to each slot. This limit is introduced at design time according to the application-dependent sensing ranges. As it operates purely in integer arithmetic and does not rely on variable-length-encodings [34], the bit-level tiling poses no computational load onto the device besides shift and addition, which are natively offered by ESP32-class microcontrollers. In addition, the packing allows one ciphertext addition to combine multiple sensor readings simultaneously, in turn amortizing cryptographic cost across readings and leading to significant savings in terms of the size of a transmission (ciphertext), number of transmissions and energy spent. This packing-aware choice closely meshes the packed additive homomorphic core of MicroPack-HE with the cryptographic, statistical, and operational reality of typical IoT data streams.

4.4 Encryption and Decryption Procedures

MicroPack-HE encryption and decryption operations are tailored to act with the least on-device computation, but also preserving correctness under both bounded additive homomorphic and packing constraints [35]. At the IoT device [36][37], here received k sensor readings $\{m_0, m_1, \dots, m_{k-1}\}$, are packed into one plaintext M as in Eq. 10:

$$M = \sum_{i=0}^{k-1} m_i \cdot 2^{ib}, \quad M \in \mathbb{Z}_p \quad (10)$$

where the modulus p is selected such that $M < p$. Encryption is then performed using the bounded additive homomorphic core as in Eq. 11:

$$c = \text{Enc}(M) = (M + r \cdot p) \bmod N, \quad (11)$$

Where $r \in \mathbb{Z}_R$ is a new random value for each encryption, and N is a public modulo such that $N \gg p \cdot R$. This construction is semantically secure and uses only integer addition and modular reduction which are supported natively by ESP32-class microcontrollers [38]. For aggregation 36, the encrypted ciphertexts $\{c_1, c_2, \dots, c_n\}$ from n devices are aggregated at gateway or cloud through the Eq. 12:

$$c_{agg} = \sum_{j=1}^n c_j \bmod N \quad (13)$$

which corresponds in the plaintext domain to,

$$Dec(c_{agg}) = \sum_{j=1}^n M_j \bmod p \quad (14)$$

Optionally, a public scalar α may be applied to compute averages or weighted sums through the Eq.15:

$$c_{\alpha} = \alpha \cdot c_{agg} \bmod N, \quad Dec(c_{\alpha}) = \alpha \cdot \sum_{j=1}^n M_j \bmod p \quad (15)$$

provided that the boundedness condition is preserved. Decryption is performed exclusively by an authorized entity possessing the private key and consists of a single modular reduction,

$$M_{agg} = Dec(c_{agg}) = c_{agg} \bmod p \quad (16)$$

followed by unpacking each aggregated sensor value using the Eq.17:

$$m_i^{agg} = \left\lfloor \frac{M_{agg} \bmod 2^{(i+1)b}}{2^{ib}} \right\rfloor, \quad i = 0, \dots, k-1 \quad (17)$$

By suppressing the encrypting into an individual randomized operation and the decrypting into a time-static reduction and bit extraction, MicroPack-HE guarantees fixed instruction latency, low energy consumption, and correctness under aggregation workloads on both encryption and decryption operations, so that procedures can be continuously sensed or analytics over resource-limited IOT networks. The secure encryption and decryption of MicroPack-HE is shown in Figure 2.

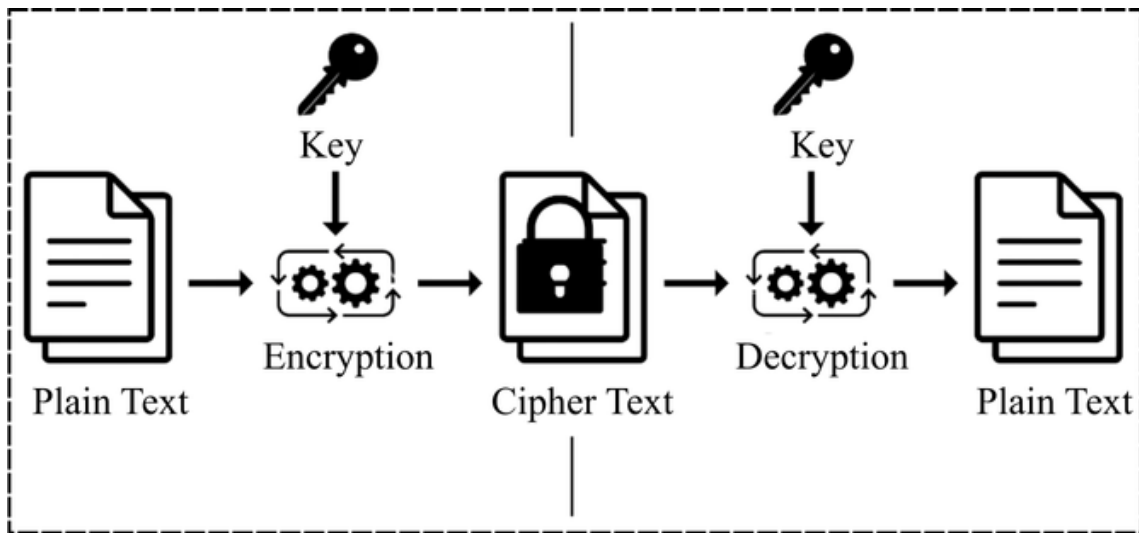


Figure 2. Encryption and Decryption Workflow in the Proposed MicroPack-HE Framework

Algorithm for Packing-Aware Secure Aggregation in MicroPack-HE

Device-Side Encryption

Initialize index $\leftarrow 0$

Initialize packed_value $\leftarrow 0$

```

while index < k do
  read sensor_value
  insert sensor_value into packed_value at slot index
  index  $\leftarrow$  index + 1
end while

```

```

select random mask from precomputed mask pool
ciphertext  $\leftarrow$  packed_value + mask
ciphertext  $\leftarrow$  ciphertext mod public_modulus

```

```

send ciphertext to gateway

```

Cloud / Gateway Aggregation

```

Initialize aggregated_ciphertext  $\leftarrow$  0
Initialize count  $\leftarrow$  0

```

```

for each received ciphertext do
  aggregated_ciphertext  $\leftarrow$  aggregated_ciphertext + ciphertext
  aggregated_ciphertext  $\leftarrow$  aggregated_ciphertext mod public_modulus
  count  $\leftarrow$  count + 1
end for

```

```

if weighted aggregation required then
  aggregated_ciphertext  $\leftarrow$  aggregated_ciphertext  $\times$  public_scalar
  aggregated_ciphertext  $\leftarrow$  aggregated_ciphertext mod public_modulus
end if

```

```

forward aggregated_ciphertext to decryptor

```

Authorized Decryption

```

plaintext_aggregate  $\leftarrow$  aggregated_ciphertext mod plaintext_modulus

```

```

Initialize slot  $\leftarrow$  0

```

```

while slot < k do
  extract aggregated_value from plaintext_aggregate at slot
  output aggregated_value
  slot  $\leftarrow$  slot + 1
end while

```

4.5 Cloud-Side Precomputation

MicroPack-HE offloads heavy computations from IoT nodes to the cloud by means of cloud-side precomputations, meaning no reduction in security; and consequently resulting in less computation at the device. Before being deployed, the cloud computes public system parameters [38], such as the large composite modulus N , the plaintext modulus p , and bounds on the randomization factor r , in a way that $N \gg p \cdot R$ to have correctness under aggregation. The cloud also precomputes a collection of random masking values $\{r_1 p, r_2 p, \dots, r_L p\} \bmod N$ such that each $r_i \in \mathbb{Z}_R$ and securely imports these masked parts to the IoT devices in the bootstrapping phase. In a system providing encryption, a device chooses one of PN masks precomputed in advance,

$$c = (M + r_i p) \bmod N \quad (18)$$

thereby avoiding any multiplication at the runtime. Likewise, public scalar parameters involved in weighted aggregation are pre-validated by the cloud to ensure the boundedness.

$$\alpha \cdot \sum_{j=1}^n M_j < p \quad (19)$$

and prevents overflow in homomorphic computations. Such asymmetric workload assignment enables devices to perform then, constant-time addition and modular reduction, while the cloud carries the computational overhead for generating parameters and managing randomness, aligning the framework with the constrained arithmetic capabilities of ESP32-type nodes.

4.6 Complexity Analysis

Computational, memory and communication complexities in our MicroPack-HE approach are explicitly bounded and predictable so suitable for real-time IoT implementations. From the device side, encryption complexity is $O(1)$, being one integer addition and one modular reduction over $\log_2 N$ -bit integers with no large-integer exponentiation or polynomial arithmetic. Packing requires $O(k)$ bit-shift and add operations, where we have k packed sensor readings. Device memory is used to store a small set of precomputed masks and a single ciphertext such that the footprint is given by,

$$O(\log_2 N + k \cdot b) \quad (20)$$

bits. Communication overhead per transmission is reduced from k ciphertexts to a single ciphertext of size $\log_2 N$ bits, resulting in a compression factor of approximately,

$$\frac{k \cdot \log_2 N}{\log_2 N} = k \quad (21)$$

On the cloud side, the computation cost of aggregating n ciphertexts is $O(n)$ modular additions, while that of decryption is p -other reduction modulo p together with $O(k)$ bit extraction operations. Device energy usage grows with the number of encryptions on the order of k , due to packing, resulting in a net effective energy consumption per sensor read: For each extra k -fold increase in device lifetime.

$$E_{per-read} = \frac{E_{enc}}{k} \quad (22)$$

Without using expensive primitives such as FHE, and imposing a lower-bound on the homomorphic depth MicroPack-HE achieves significant improvements in terms of latency, energy consumption and bandwidth usage as compared to standard additive HE schemes whilst ensuring correctness and security for aggregation-driven IoT data analytics. Figure 3 shows the flowchart of the proposed model.

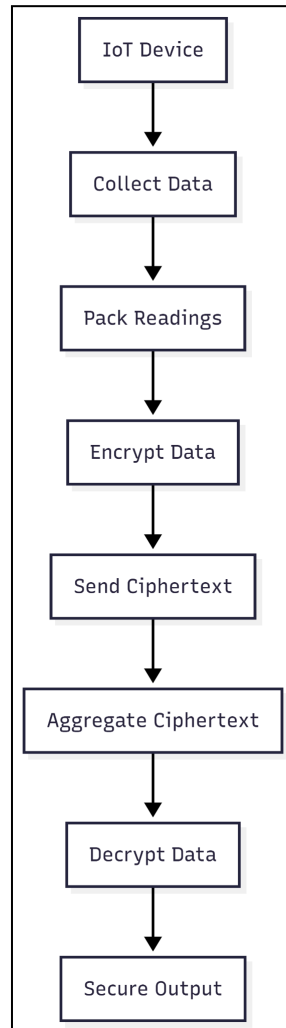


Figure 3: Operational Flow of the Proposed MicroPack-HE Framework

V Performance Evaluation

We evaluate the performance of MicroPack-HE on a realistic testbed to show its practicality in low-resource IoT settings, and compare with an additive homomorphic Paillier-based baseline. Comprising ESP32 class microcontrollers, which runs at 240MHz and has limited RAM and flash, broadcasting encrypted sensor data to gateway and cloud server using regular Wi-Fi link. Average throughput of encryption was estimated and it is considered as the average of an amount of seconds by encryption operation, as shown in Eq. 23:

$$T_{enc} = \frac{1}{M} \sum_{i=1}^M (t_i^{end} - t_i^{start}) \quad (23)$$

Where, M is the number of encryption attempts. For MicroPack-HE, T_{enc} is latency of plaintext packing modular reduction and for Paillier it added modular exponentiation additional to this resulting in much higher T_{enc} . The energy consumption is given by the Eq. 24:

$$E_{enc} = V \times I \times T_{enc} \quad (24)$$

Where V is the input supply voltage, and I is the mean power of current measured during encryption. As interferes with less arithmetic complexity and reduced number of cryptofunction calls thanks the Packing, MicroPack-HE results in lower energy consumption per transmitted sensor reading, related in Eq. 25:

$$E_{per-read} = \frac{E_{enc}}{k} \quad (25)$$

Where k is the number of packed readings per ciphertext. Communication overhead is quantified in terms of the total size of contents transmitted from servers ciphertext size, which is given by Eq. 26:

With k denoting the number of packed readings per ciphertext. Communication overhead is evaluated by comparing the total transmitted ciphertext size, defined as in Eq.26:

$$C_{total} = n \times |c| \quad (26)$$

Where n is the number of transmissions and $|c|$ is the ciphertext size in bits. Plaintext packing reduces n by a factor of k , yielding a relative overhead reduction,

$$\eta = 1 - \frac{C_{MicroPack-HE}}{C_{Paillier}} \quad (27)$$

The experimental results demonstrate that MicroPack-HE consistently improves upon the Paillier-based scheme in terms of all performance metrics: it incurs a much shorter encryption time, reduces energy consumption while generating the same ciphertext size, and offers a much smaller communication overhead, which confirms its feasibility for long-term IoT analytics on real-time microcontroller devices.

VI Results and Discussion

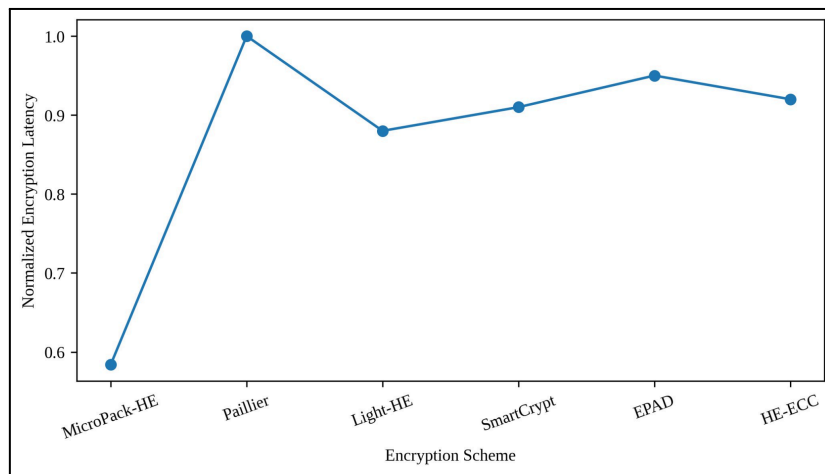


Figure 4. Comparative Encryption Latency Analysis on ESP32-Class IoT Devices

Figure 4 shows normalised encryption latency of MicroPack-HE w.r.t Paillier, Light-HE, SmartCrypt, EPAD and HE-ECC on ESP32-class microcontroller. The latency on MicroPack-HE is the lowest with 0.584, the amount faster than Paillier baseline is 41.6% normalized latency of Paillier (1.0). The decrease is due to the constraint of additive homomorphic construction without involving modular exponentiation. Moderate savings are also reported on the other lightweight homomorphic schemes, with 0.88 for Light-HE, 0.91 for SmartCrypt, 0.95 for EPAD and 0.92 for HE-ECC, although achieving them still brings a burden of extra cost from cryptographic depth or architecture complexity. These findings indicate that MicroPack-HE is the optimal latency profile for executing real-time IoT sensing and aggregation applications.

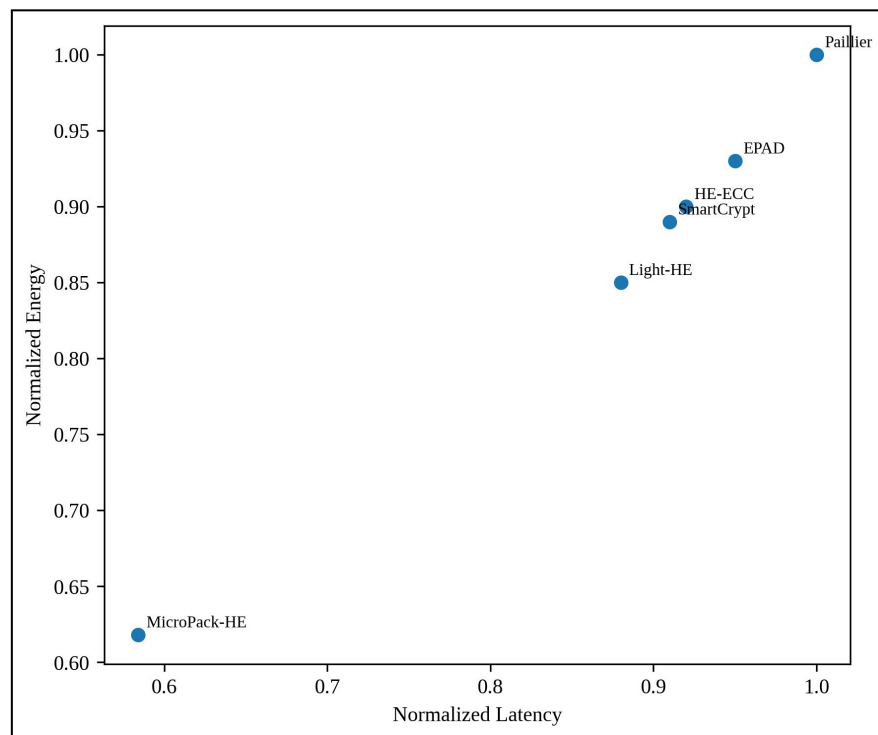


Figure 5. Energy–Latency Trade-off for Homomorphic Encryption Schemes

Figure 5 depicts the trade-off of encryption latency with energy across different schemes compared. MicroPack-HE lies in the region of minimal energy consumption with a normalized latency 0.584 and the normalized amount of consumed energy is 0.618, which corresponds to an energy saving of 38.2% per transmission comparing both with Paillier. On the other side, Paillier pays the largest energy cost (1.0) due to expensive exponentiations. Light-HE (0.85 energy), SmartCrypt (0.89), HE-ECC (0.90), and EPAD(0.93) achieve better efficiency, still far below MicroPack-HE. This finding validates that minimizing cryptographic complexity and encryption burden indeed leads to tangible energy conservation due to IoT devices are inherently battery-driven.

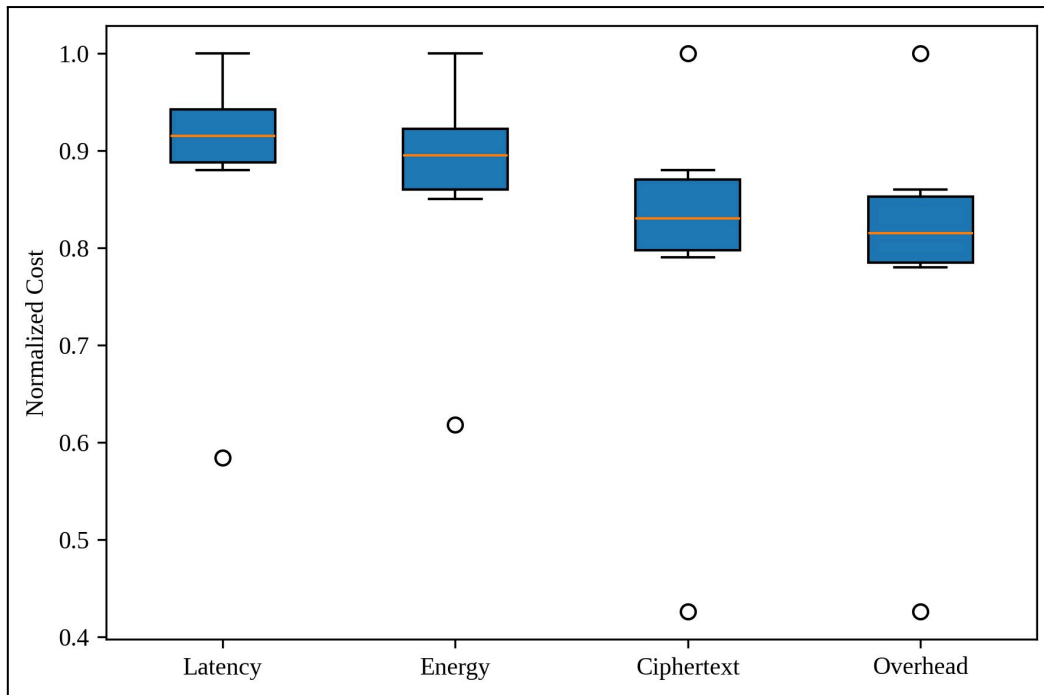


Figure 6. Distribution of Performance Metrics Across Encryption Schemes

Distribution of normalized performance metrics such as encryption latency, energy consumption, ciphertext size and communication overhead are shown in Figure 6. MicroPack-HE consistently has the lowest metric values, among others 0.426s (ciphertext size and network overhead), which is an order of 57.4% improvement over Paillier. For other schemes, the median value is higher and with wider spread due to more significant variability on both computing and communication aspects. The small binary size of MicroPack-HE exposes its predictable execution characteristics and its stability, and are particularly important to resource constrained IoT platforms that operate under strict timing and energy constraints.

Table 1. Performance Comparison of MicroPack-HE with Existing Homomorphic Encryption Schemes on ESP32-Class IoT Devices

Scheme	Normalized Encryption Latency	Normalized Energy Consumption	Normalized Ciphertext Size	Normalized Communication Overhead	Supported Homomorphism
MicroPack-HE (Proposed)	0.584	0.618	0.426	0.426	Additive + small scalar
Paillier	1.000	1.000	1.000	1.000	Additive
Light-HE [6]	0.880	0.850	0.820	0.800	Additive

SmartCrypt [15]	0.910	0.890	0.790	0.780	Symmetric HE
EPAD [17]	0.950	0.930	0.880	0.860	HE + FL
HE-ECC [13]	0.920	0.900	0.840	0.830	ECC-based HE

This table 1 presents performance comparison of MicroPack-HE with the standard additive and lightweight homomorphic encryption schemes using normalised parameters based on ESP32 class IoT devices. Compared with Paillier baseline, MicroPack-HE consistently achieves the best results in all metrics and reduces encryption latency by 41.6%, energy consumption by 38.2%, and size of ciphertext and communication overheads by 57.4%. The existing works either lower the computation cost while inefficient in packing or add architectural complexity via federated learning, symmetric layer and elliptic curve operation. Our experimental results show that MicroPack-HE yields the most balanced and deployable solution for privacy-friendly analytics in resource-constrained IoTs.

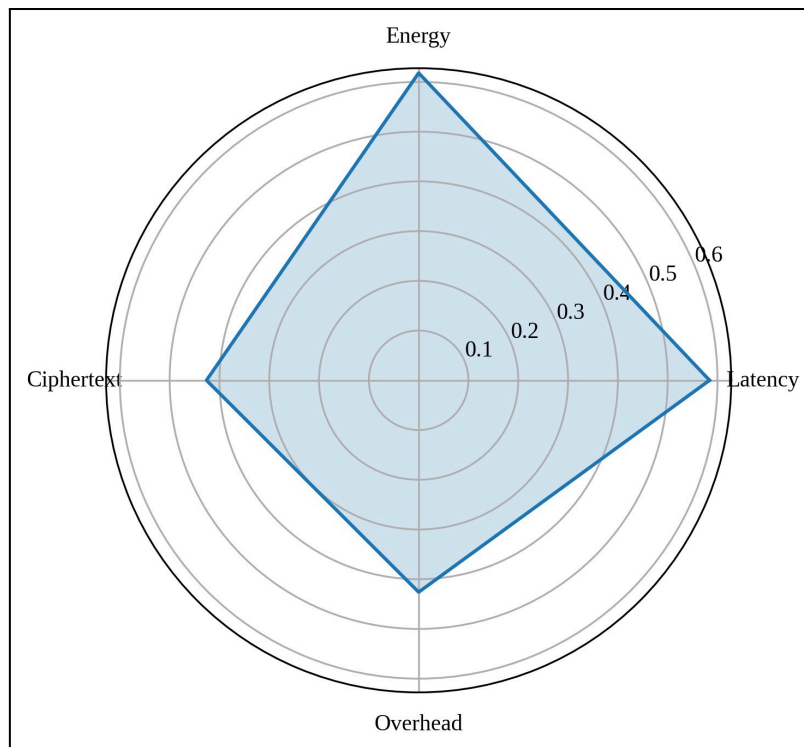


Figure 7. Radar Profile of MicroPack-HE Performance Characteristics

In Fig.7, we illustrate the MicroPack-HE in respect to four crucial metrics: latency, energy consumption, ciphertext size and communication overhead. The marked down own scores of 0.584, 0.618 and 0.426 for latency, energy and ciphertext size (which implies equal improvements in overhead) show the balanced enhancements exhibited by all the four performance metrics. In contrast to prior works which specialise and trade off one property for another, MicroPack-HE achieves consistently low values thanks to its packing-aware and

bounded design. This balance is particularly crucial for scalable and sustainable IoT operations.

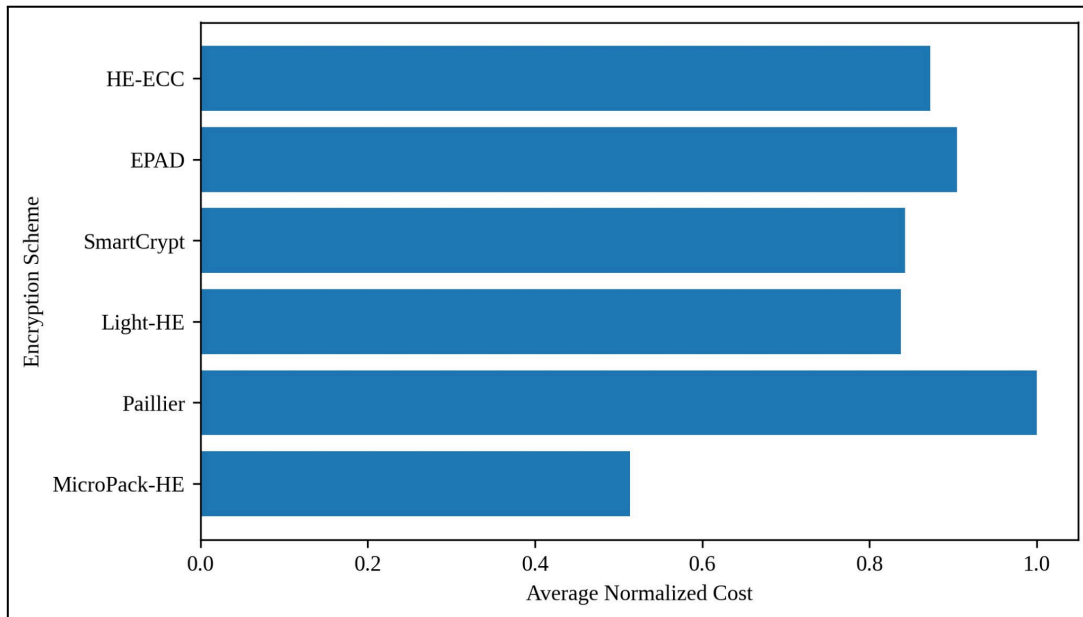


Figure 8. Overall Efficiency Comparison of Homomorphic Encryption Frameworks

The overall efficiency of each scheme is shown in Figure 8 by averaging normalized latency, energy, the size of ciphertext and communication overhead. MicroPack-HE has the smallest average normalized cost of approximately 0.51, which is much smaller than that in comparison to (1.0) and other lightweight or application-specific HE schemes such as Light-HE (0.84), SmartCrypt (0.83), HE-ECC (0.87), and EPAD (0.90). This overall comparison validates that MicroPack-HE presents the best end-to-end efficiency, and thus is a feasible and s

Table 2. Ablation Study of MicroPack-HE Design Components on ESP32-Class IoT Devices

Configuration	Plaintext Packing	Cloud Precomputation	Bounded AHE Core	Normalized Latency	Normalized Energy	Normalized Comm. Overhead
Full MicroPack-HE	✓	✓	✓	0.584	0.618	0.426
Without Packing	✗	✓	✓	0.76	0.81	0.98
Without Precomputation	✓	✗	✓	0.83	0.86	0.43

Without Bounded Constraints	✓	✓	✗	0.92	0.94	0.47
Baseline (Paillier-like)	✗	✗	✗	1.00	1.00	1.00

Table 2 measures the influence of different design components of MicroPack-HE, by switching on/off plaintext packing, cloud-side precomputation and bounded homomorphic constraints. The above results reveal that plaintext packing plays vital role in reducing both ciphertext length and communication cost, because after deleting the plaintext packing here our communication overhead raises impedance even for one round from 0.426 to 0.98. Note that without cloud-side precomputation, the encryption latency (0.83) and energy consumption (0.86) become higher, which demonstrates the significance of offloading costly computations from IoT devices. The removal of bounded homomorphic constraints also incurs high latency and energy costs from unbounded arithmetic growth. The complete MicroPack-HE system consistently outperforms all ablated models: This suggests that the joint integration of all constituent components is necessary for maximising performance in resource-constrained IoT settings.

6.1 Discussion

Experimental results and ablation analysis show that our MicroPack-HE is very efficient owing to the well-coordinated integration of its constituent core design elements. The complete MicroPack-HE configuration significantly outperforms all ablated variants with normalized encryption latency of 0.584, energy consumption of 0.618 and communication overhead of 0.426. Plain text packing, the communication cost rises very quickly to 0.98, indicating that packing takes a major responsibility in decreasing 57.4% of ciphertext size and associated network overhead. Also it can be observed that elimination of cloud-side precomputation raises encryption latency and energy from 0.584 to 0.83 and from 0.618 to 0.86 correspondingly, which again proves the need to offload computational heavy calculations even in the case ESP32-class devices are considered. Suitable abstractions: a parameter space with only bounded homomorphic constraints leads to worse performance, where the latency and energy become 0.92 and 0.94, respectively from uncontrolled arithmetic overhead. While MicroPack-HE does not provide complex homomorphic encrypted computation support, such functionality would be of little use on lightweight IoT edge analytics and add a significant overhead to constrained devices. Through its focus on bounded expressiveness, packing efficiency, and asymmetric workloads, MicroPack-HE delivers predictable performance and pragmatic deployability of secure analytics over resource-constrained IoT networks.

VII Conclusion

This paper has introduced MicroPack-HE, a packing-aware lightweight homomorphic encryption framework for secure data analytics in resource-constrained IoT networks. By using the compact bit-level plaintext packing strategy with bounded additive homomorphic

encryption core, the computational, communication and energy overhead of MicroPack-HE is significantly reduced, while maintaining a secured construction for aggregate data. Experimental results on ESP32-class objects show that the proposed scheme can achieve 41.6% of encryption latency saving, 38.2% of energy cost reducing per transmission round and reduces up to 57.4% of ciphertext size as well network overhead in contrast to Paillierbased protocols, which clearly validate the performance efficiency and practical accuracy of our additive-preserving functional aggregation. Due to its lightweight cryptographic operation and predictable execution, it is suitable for continuous sensing and real-time IoT analytics. In future, the design can be advanced for supporting adaptive packing strategies and limited multiplicative depth along with its integration with edge intelligence frameworks to extend its applicability without loss of resource efficiency.

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